

Decentralized management of intersections of automated guided vehicles

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Abstract: The emerging topic of cooperative intersection management for vehicles has raised up new solutions for traffic control in order to avoid collisions, deadlocks, and improve traffic efficiency. The solutions developed for road traffic can easily be applied to automated guided vehicles to overcome the common drawbacks, and improve the number of vehicles in a network. In the context of a network of vehicles only regulated at intersections, we propose an algorithm in order to prevent deadlock at intersections in a network of automated guided vehicles.

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1. INTRODUCTION

In a network with a large number of automated guided vehicles (AGV), traffic control is crucial to the system performance. The traffic control must avoid collisions, deadlocks/gridlocks and ensure that every AGV can reach its destination, and for this purpose different solutions are proposed:

- To avoid collisions, zone control with wireless transmission is the favorite traffic control of most environments as it is simple to install and expand.
- To avoid deadlocks/gridlocks in the network of AGV, the mainly used solutions are improved banker's algorithms (Lawley et al., 1998; Ezpeleta et al., 2002; Bobanac and Bogdan, 2008), and the use of Petri nets (Wu and Zhou, 2007), but they both suffer from drawbacks. The banker's algorithm can eliminate valid solutions, significantly compromising the efficiency of the management, while Petri nets can cause livelocks. Moreover the number of AGV in the network with these solutions is limited: a resource is an arc between two nodes and can only be held by one AGV at a time. Other solutions relying on siphon-based deadlock prevention are optimal but the minimal siphon is a NP problem, thus it cannot be considered for real time applications.
- Conflict-free routing based solutions are also proposed in Nishi and Tanaka (2012) and Miyamoto and Inoue (2016), they prevent the intersection problem but they require a centralized management, and some valid solutions are eliminated.

Due to the recent development of cooperative intersection management for road vehicles, new solutions have emerged to improve traffic efficiency with both collision and deadlock avoidance. One can quote the well-known Reservation-Based Protocol (RBP) where the vehicle sends a reservation request of space and time to the

server in Dresner and Stone (2004) and de La Fortelle (2010). Other works are based on Cooperative Adaptive Cruise Control at Intersections (CACCI). The server detects conflicts and accordingly sends acceleration and deceleration messages to vehicles in order to avoid collisions in Zohdy and Rakha (2012) and Zohdy et al. (2012). Furthermore a Sequence-Based Protocol (SBP) is proposed in Perronnet et al. (2013). It assumes that the intersection is controlled as follows : either the intersection manager or a decentralized negotiation explicitly determines the sequence of vehicles in each conflict area. The sequence determines which vehicle is the first, which one is the second and so on.

Though these protocols are designed for isolated intersection, they can be integrated in a more global solution in charge of managing (routing and gridlock prevention) a fleet of vehicles, like Bocewicz et al. (2007) and Perronnet et al. (2014). For instance, in Perronnet et al. (2014) a protocol is proposed to avoid gridlock (deadlock caused by the interaction of multiple intersections) in a network of intersections allowing more than one vehicle per zone between intersections. This algorithm is based on the principle of path reservation: a vehicle has to follow a given path, it asks the global server for the authorization to follow this path, then the global server reserves the path of this vehicle and informs intersection servers. Then, eventual intersections of vehicles are locally managed by one of the previously presented solution with respect to the constraints stated by the global protocol.

These solutions are originally designed for road vehicles but can be applied to AGV in order to increase the number of AGV in a network, and improve the overall performance. Therefore, in the context of this paper, we assume a network with a traffic control protocol in charge of the gridlock prevention allowing more than one vehicle between two intersections. We then consider an intersection in this network with multiple unlocked vehicles, i.e. vehicles that

can move without causing a gridlock. In order to avoid collisions between AGV, we assume that the intersection is regulated with an intersection server using a SBP and a first-in, first-out scheduling policy.

According to the chosen protocol, the inconsistency of the presence list brings different risks. Deadlock and collision are then possibles. In the following we will consider a centralized architecture of SBP (C-SBP) named Transparent Intersection Manager (TIM) (Perronnet et al., 2013). In TIM vehicles synchronize their speeds according to the presence list received from the server. There are two advantages of C-SBP. The first one is the default-deny: if a vehicle is not able to establish a communication with the server, it has to brake before the box junction. The second one is that the results of C-SBP can easily be extended to RBP.

However, due to potential communication problems (message losses), the sequence built according to the order of arrival of messages from the AGV can be different from the physical order of the vehicles, resulting in a deadlock situation. The scope of this paper is to propose a re-sequencing algorithm able to avoid the deadlock without introducing any risk of collision (3), even with an unreliable communication. In order to assess the proposed algorithm, simulations and intersections of robots are performed.

This paper is organized as follows; first it presents the protocol TIM and the conditions of the problem. Then, the paper introduces the deadlock problem as well as the collision risk due to a bad re-sequencing. Therefore, it presents the re-sequencing algorithm and shows that the resulting sequence is collision-free. Before concluding, the paper discuss the results of simulations as well as the results of the cooperative intersection of robots.

2. PROBLEM STATEMENT

2.1 Brief presentation of the Transparent Intersection Manager

Vehicles in TIM are able to observe the obstacles in the surrounding environment and to adapt their speed accordingly. AGV move autonomously if all obstacles are visible by the sensors. In order to enhance the intersection safety and throughput, a negotiation protocol based on wireless communication to synchronize the AGVs speed is proposed by TIM. Every AGV has to use wireless communication to inform an intersection server about its movement. More precisely, every AGV communicates its current position and speed as well as the desired destination and the remaining distance before the exit of the box junction. Accordingly, the intersection server broadcasts these data through an ordered presence list to all the AGV closed to the intersection. In the received presence list, the first vehicle has the highest priority and so on. Each AGV considers three kinds of obstacles:

- Visible obstacles: mainly a precedent AGV in the same buffer lane. Visible obstacles are detected by the sensors without any message received from the server.
- Virtual obstacles: conflicting AGV with higher priority.

- Stop line: the nearest border of the box junction. For safety reason, the AGV has to stop by default before the stop line (default-deny principle).

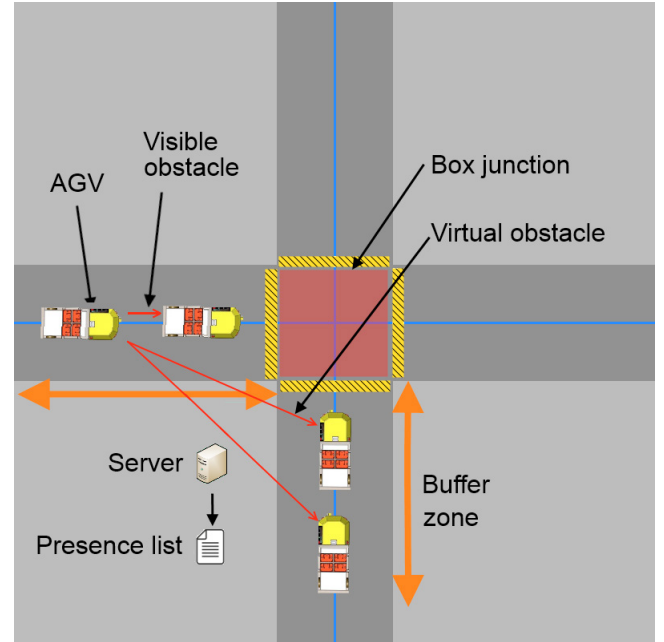


Fig. 1. TIM

The three obstacles are presented in 1. For each obstacle, the AGV computes an acceleration. Hence, there are three computed accelerations:

- a_r to keep a safe distance from the visible obstacles.
- a_i to keep a safe distance from priority conflicting
- a_s to allow a vehicle to stop in the case of a dangerous situation.

The three accelerations contribute to determine the acceleration of the AGV as follows :

- a_i is determined as the minimum acceleration from all conflicting vehicles with a higher priority.
- If the AGV has not received a presence list it has to stop before the box junction. In other words, the intersection map is known before requesting the presence list.
- Vehicles are not allowed to overtake in the buffer zone of the intersection.
- The speed synchronization is done near the junction box.

Formally, the acceleration of each AGV is computed as $a = \min(a_r, \max(a_i, a_s))$

We draw the reader attention to the fact that only a_i depends on the presence list sent by the server. The vehicle must know the intersection map before getting into the buffer zone. We highlight also the fact that, default deny (a_s) is used when the vehicle has not yet received a presence list or when it is not in the presence list. It is also used when the vehicle is not able to keep a safe distance from the virtual obstacles.

2.2 Considered assumptions

TIM allows several strategies for synchronizing the speed of vehicles as well as many policies for ordering the presence list. In the reminder we consider the following assumptions:

- The error on the position is below one meter.
- With respect to the positioning accuracy, the positions received by the manager can refer to old positions of the vehicles, but never to a position that the vehicle has not reached yet. In other words, more than one meter error is due communication delays between the AGV and the intersection server.
- a_i is computed under the constraint that two conflicting vehicles cannot simultaneously access to the box junction.
- When $t \rightarrow \infty$, all present vehicles will be discovered at the good position.
- The presence list is built according to "first-come, first-served" (FCFS).

2.3 Loss of packets

Using wireless communication, latency and loss of packets are common problems. And as the number of communicating AGV increases in a network, they become a major preoccupation. Therefore the order of arrival of vehicles at the intersection can differ from the order of arrival of the messages notifying their arrival.

The default-deny allows better safety conditions with the intersection. Indeed, if an AGV is not able to communicate with other AGV through the intersection server, it has to stop before the box junction. As long as an AGV is not in the presence list sent by the server, it has to stop. However, in such a case the default deny can block the traffic (Lombard et al., 2015). More precisely, if one vehicle stops then all vehicles behind it in the same buffer zone will stop too. Furthermore all conflicting AGV with a lower priority will stop. The main raised issue is how to safely recover the traffic movement when a missed vehicle establishes a connection.

As said before the focus is put on the FCFS policy. With this policy the first AGV arrived at the buffer zone will be the first allowed to cross the conflict area according to the presence list. When considering an instantaneous wireless communication, the order of arrival of vehicles is defined by the order of arrival of the messages emitted by the AGV toward the intersection server.

Let us consider the example presented in 2. AGV are numbered according to the messages received by the server. One can note that 10 and 11 were initially missed. Because vehicle 10 is missed, only vehicles 1 and 4 are able to cross the intersection. If the intersection server and the AGV operate under normal conditions, the server will gradually discover the missed vehicles. When these vehicles will be discovered, the server will have to include them in the presence list.

If vehicles are ordered according to the received message there is a deadlock. Indeed, there is a circular wait: $10 \succ 9 \succ 7 \succ (6,5) \succ (3,4) \succ 2 \succ 10$ ($i \succ j$ means that i is waiting for j to cross the intersection). Only AGV #1

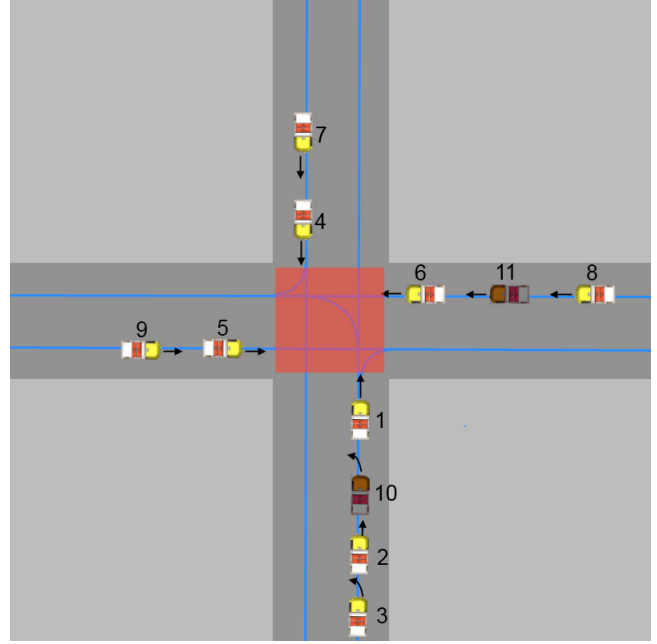


Fig. 2. Two missed vehicles: The number associated to vehicles shows the order of the received messages by intersection server.

and #4 are able to cross the intersection. If the AGV are ordered according to their arrival time, a collision may happen. Indeed, AGV #4 discovers a new virtual obstacle (#10) that has not been initially considered for keeping a safe distance. Moreover, if we consider that the AGV #11 has arrived before AGV #7, the latter has not adapted its speed to avoid collision with #11. Hence, the main raised issue is to reconsider the presence list with missed vehicles without deadlock and collision risks.

3. APPROACH FOR SAFE COOPERATIVE INTERSECTION MANAGEMENT

To achieve the objective, in the following we will provide an algorithm that first prevents deadlock and then avoids collision.

3.1 Collision free

First let us draw the reader attention to the fact that when a new vehicle is discovered, if it is not ordered correctly a collision may happen, whether or not a cycle is discovered. Indeed, let us consider the example given in 2. If only AGV #1, #2, #3 and #4 are present and if we discover vehicle #10, there is no circular wait, but there is a risk of collision between #10 and #4. This is also the case of vehicle 11 for which there is no cycle. In Qian et al. (2014), the authors introduced and proved the collision free condition. There is a collision risk if a vehicle discovers a new obstacle. From this condition we address a conflict vector of the new discovered vehicle with all present vehicles and we proceed as follows:

- The new discovered vehicle is ordered after the last conflicting vehicle.
- Each AGV in the same lane physically behind the new discovered AGV move back in the list until it is behind the vehicle physically in front.

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function SIZEOF(s)           ▷ size of the collection s

function L(v)                 ▷ lane of the vehicle v

function D(v)                 ▷ distance of the vehicle v from the intersection

function S(v)                 ▷ position of the vehicle v in the sequence

function CONFLICT(v1, v2)    ▷ true if v1 and v2 are in conflict,
false otherwise

procedure MOVE(s, i, j)      ▷ move the element at position i in the
sequence s to position j

procedure INSERT(s, i, o)    ▷ insert the object o at position i in s
procedure INSERTIONSEQUENCE(v)
  expectedPosition ← sizeof(sequence)
  for all v' in sequence do
    if l(v) = l(v') and d(v) < d(v') then
      expectedPosition ← min(expectedPosition, s(v'))
    end if
  end for
  insertionPosition ← expectedPosition
  if expectedPosition ≠ sizeof(sequence) then
    for i from expectedPosition to sizeof(sequence) - 1 do
      if conflict(v, sequence[i]) then
        insertionPosition ← i
      end if
    end for
    moved, i ← 0
    while i < insertionPosition - moved do
      if l(v) = l(sequence[i]) then
        MOVE(sequence, i, insertionPosition)
        moved ← moved + 1
      else
        i ← i + 1
      end if
    end while
    insertionPosition ← insertionPosition - moved
  end if
  INSERT(sequence, insertionPosition, v)
end procedure

```

Fig. 3. Collision and deadlock free algorithm (CDF algorithm)

The optimized algorithm called CDF is given in 3. The algorithm is collision free since there is no vehicle which will discover new obstacle. Indeed, first, the conflicting vehicles with the new discovered vehicle will all pass before. Second, the vehicles behind can only loose priority. It remains to show that it is deadlock free.

3.2 Recovering liveness

There are four conditions to have a deadlock situation Coffman et al. (1971). If they are satisfied simultaneously, a deadlock takes place:

- Mutual exclusion
- Hold and wait
- No preemption
- Circular wait

The three first conditions are inherent to the traffic. Hence, when there is a circular wait, a deadlock happens. The circular wait can be represented as a directed cycle in the graph of dependency. To recover liveness, we need to

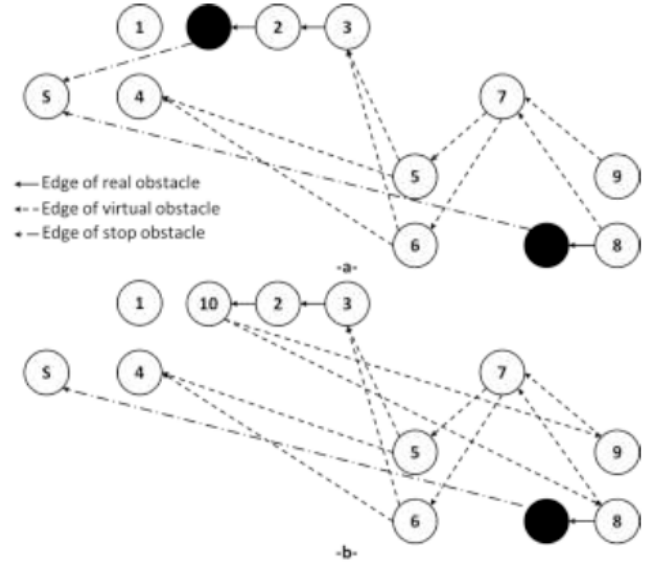


Fig. 4. Graph representation of the intersection: a- before receiving message from vehicle 10, b- after receiving message from vehicle 10

break the cycle by removing edges. However, only virtual obstacles can be reconsidered.

Let $G = (N, E)$ be a directed graph of dependency in which the set of nodes $N = 1, 2, \dots, N \cup S$ represents the vehicles with the stop constraint and the set of directed edges E connects each node v to node v' if the vehicle v has to adapt its speed according to the obstacle v' . Hence, $E = E_r \cup E_i \cup E_s$ according to the minimal acceleration used to cross the intersection. 4-a and 4-b show the graph before and after discovering vehicle 10, respectively. For the sake of readability, edges of real obstacles are removed in 4 if the order is respected by edges of virtual obstacles. One can note that in both situations there is no collision risk because vehicles are stopped, waiting for vehicle 10. In 4-b there are four cycles given in 5. The cycles can be broken by removing two edges of virtual constraints. There are four possible solutions:

- $10 \succ 9$ and $10 \succ 8$,
- $9 \succ 7$ and $8 \succ 7$,
- $7 \succ 5$ and $7 \succ 6$,
- $5 \succ 3$ and $6 \succ 3$.

From the CDF algorithm presented in 3, edges $5 \succ 3$ and $6 \succ 3$ are removed. The new sequence when vehicle 10 is discovered is $1 \prec 4 \prec 5 \prec 6 \prec 7 \prec 8 \prec 9 \prec 10 \prec 2 \prec 3$. When vehicle 11 is discovered, the new sequence will be $1 \prec 4 \prec 5 \prec 6 \prec 7 \prec 9 \prec 10 \prec 2 \prec 3 \prec 11 \prec 8$

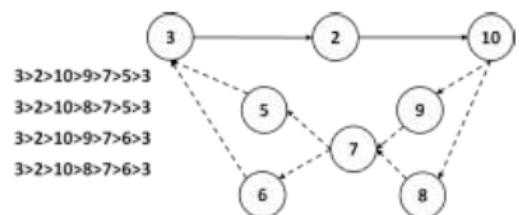


Fig. 5. The four circular wait

It can be shown that there is a circular wait if the two following conditions are satisfied:

- The set S of vehicles physically behind the new discovered vehicle in the same lane is nonempty
- There is at least one vehicle in S which precedes a conflicting vehicle with the new discovered vehicle

Then, since all vehicles physically behind the new discovered vehicle are ordered after, the link that exists with a conflicting vehicle is removed. Hence the cycle is interrupted.

4. TESTS AND RESULTS

4.1 Simulation framework

Simulations have been realized to experiment this solution. They were realized in Java with a simple yet realistic AGV model, introducing inaccuracy and delays in the positioning system, in the front radar measure, and delays in the application of the acceleration and direction command. In order to test our deadlock solution, AGV are placed on an 8-shaped circuit where the crossing point is the intersection. The vehicles use the TIM protocol (Perronnet et al., 2013) in order to get the right-of-way at the intersection.

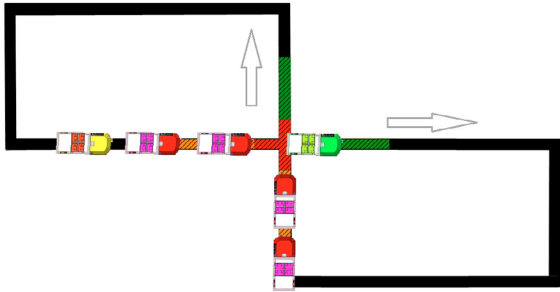


Fig. 6. Illustration of the simulation circuit

A communication rate has been fixed to 0.5s and different loss rates have been tested, in order to simulate the issues of a wireless communication. To represent the loss rate, when a message is sent (either by the vehicle or the intersection manager), it has a fixed probability to be ignored. The simulations have been realized using the loss rates 5%, 10%, 50%, 70%, 90% and 95%, with and without the deadlock prevention.

The reader may note that these loss rates do not reproduce accurately the loss rates of a V2I communication, but are used to stress-test our solution in critical conditions (if you can do the big things, you can do the little things as well). Moreover the presentation of the number of vehicles evacuated is used to show the liveness or the deadlock situation of the intersection, but is not used here to evaluate the throughput of the intersection using TIM as it has already been studied in Perronnet et al. (2013).

4.2 Results

The following figures (7 and 8) represents the number of vehicles crossing the intersection of the circuit in function of the time for different loss rates. These measures have been made with six vehicles on the circuit. On 7, we can

observe a deadlock situation after 880 seconds for a loss rate of 70%, 380 seconds for a loss rate of 95%, and 38 seconds for a loss rate of 90%. Even if is not represented here, we have observed that a smaller amount of lost packets can also lead to a deadlock situation after enough time. These statements confirm our assumption that, even if in real conditions the risk is low, deadlock cannot be ignored.

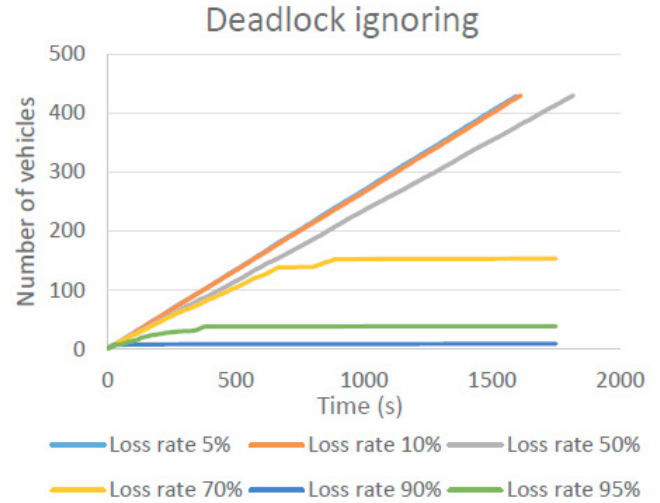


Fig. 7. Evacuation of vehicles without deadlock solution

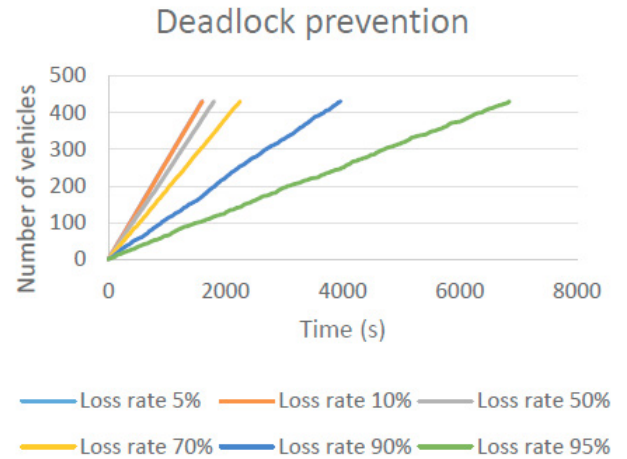


Fig. 8. Evacuation of vehicles with deadlock prevention

As it can be seen on 8, with deadlock prevention vehicles were sometime stopped at the intersection due to the loss of messages (if they do not receive the right-of-way from the manager, their default behavior is to stop), but they always ended up moving out of the intersection. Moreover, no collisions were detected: a hazardous situation is counted every time the distance between two vehicles is below one meter, this did not happen in our simulation.

4.3 Tests with EV3 robots

Finally, to confirm the feasibility of the solution, and its collision-free property, real tests have been made using Lego EV3 robots. Our test bench is presented in 9.

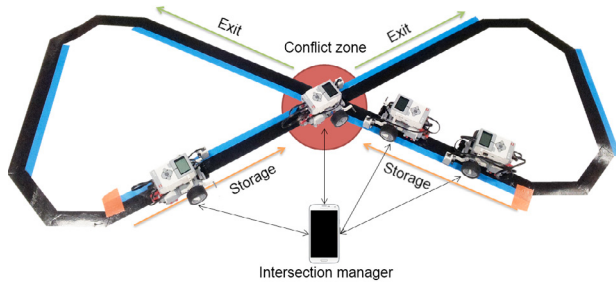


Fig. 9. EV3 intersection test bench

Positioning of the robots was made using the odometry, the color sensor and the color marks on the ground, the front obstacle detection using the EV3 sonar and the network using Edimax Wi-Fi dongles. The intersection manager was an Android phone acting as a wireless access point for the robots. As expected, during first tests, no collision neither deadlocks were observed, but the quality of the wireless connection was good (only five devices were using Wi-Fi and they were close to each other). A fake network disturbance has been introduced similar to the one used in simulations in order to stress-test our protocol. It consisted of a fixed probability that the intersection manager ignores a received message. It slowed down the performance of the intersection but collisions and deadlocks were not introduced.

5. CONCLUSION

During our work to extend the tests on cooperative intersection management to three vehicles and more, we have faced deadlock situations. The purpose of this paper has been twofold. In a first part, it has been shown that the impact of vehicular communication issues (message delays and losses) can lead to a specific case of deadlock resulting of the interdependency between the right-of-way and the position of a vehicle in its lane. In a second part, a solution to prevent deadlocks has been presented, and it has been shown that the proposed solution will effectively prevent deadlock and will not provoke collisions.

Finally, simulations and tests have been realized. They showed up that the presented kind of deadlock cannot be ignored, especially when the message loss rate increases. Moreover, it confirmed that the proposed solution prevents the risk of deadlock and do not provoke any collision.

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